

🏠 Домашний Шеф — Quick Reference Card

📦 PRODUCT MASTER SYSTEM

Product ID Format

[CATEGORY] - [SUBCATEGORY] - [NUMBER]

ML-CR-001 ← Молочные > Сливки > №1

VG-TM-002 ← Овощи > Томаты > №2

MT-CN-003 ← Мясо > Курица > №3

Categories & Codes

Код	Категория	Подкатегории
ML	Молочные 🥛	CR(Сливки), YG(Йогурт), CH(Сыр), MK(Молоко)
VG	Овощи 🥬	TM(Томаты), PP(Перец), ON(Лук), GB(Зелень)
MT	Мясо & Рыба 🍖	CH(Курица), BF(Говядина), FH(Рыба), SF(Морепродукты)
PA	Панир & Основы 🥘	OL(Масла), BR(Крупы), PS(Паста), SL(Соль & Специи)
DK	Напитки 🥤	WT(Вода), CF(Кофе), TE(Чай)
FR	Фрукты 🍏	AP(Яблоки), OR(Апельсины), BR(Ягоды)

Product Master Fields (Minimum)

```
{
  "productId": "ML-CR-001",
  "russianName": "Сливки 33%",
  "englishName": "Cream 33%",
  "originalName": "Slagroom",
  "category": "ML",
  "subcategory": "CR",
  "baseUnit": "ml",
```

```
"storageType": "Refrigerated",
"shelfLife": 10,
"openShelfLife": 3,
"minimumStock": 500,
"averagePrice": 2.49,
"defaultCurrency": "EUR"
}
```

✅ CRITICAL RULES

- ❌ **NEVER** change productId
 - ✅ Everything else is mutable
 - 🔗 All systems link through productId
 - 🌐 One productId = one base product (regardless of brand)
-

📊 INVENTORY SYSTEM

Separate from Product Master!

```
Product Master    = WHAT & HOW
Inventory          = WHEN & WHERE & HOW MANY
```

Inventory Fields

```
{
  "inventoryId": "INV-001",
  "productId": "ML-CR-001",
  "quantity": 500,
  "unit": "ml",

  "location": "Refrigerator",
  "purchaseDate": "2024-01-15",
  "expirationDate": "2024-01-25",
  "openDate": null,

  "status": "Fresh",
  "costPerUnit": 2.49,
  "currency": "EUR"
}
```

Storage Locations

	Refrigerator	(0-4°C)
	Freezer	(-18°C)
	Pantry	(Room Temp)
	Prep	(Active ingredients)
	Ready Meals	(Prepared dishes)

Inventory Statuses

Fresh	→ New, unopened
Opened	→ In use
Frozen	→ In freezer
Defrosting	→ Thawing process
Defrosted	→ Fully thawed
Expiring Soon	→ Days remaining < 2
Expired	→ Past expiration

Status Transitions

Fresh
└→ Opened (when used)
└→ Frozen (moved to freezer)
└→ Expiring Soon → Expired

Frozen → Defrosting → Defrosted → (use or discard)

RECEIPT SYSTEM (Чеки)

Workflow

1. Scan Receipt (photo)
↓
2. OCR Recognition
↓
3. Parse Items (Dutch)
↓
4. Translate Dutch→Russian
↓
5. Match to Product Master

- ↓
6. Auto-add to Inventory
- ↓
7. Convert EUR→RUB (if needed)

Dutch → Russian Mapping

```
{
  "slagroom": "Сливки",
  "tomaten": "Томаты",
  "kipfilet": "Куриное филе",
  "kaas": "Сыр",
  "melk": "Молоко",
  "olijfolie": "Оливковое масло",
  "ui": "Лук",
  "knoflook": "Чеснок",
  "paprika": "Перец"
}
```

Currency Conversion

EUR → RUB (rate ≈ 100)

Example: 2.49 EUR = ~249 RUB

RECIPES SYSTEM

Recipe Points to Product ID (NOT specific product)

```
//  CORRECT
ingredient: { productId: "ML-CR-001", quantity: 500, unit: "ml" }

//  WRONG
ingredient: { productName: "Сливки", quantity: 500 }
```

Why?

If recipe references "ML-CR-001" (Сливки):

- Can use any brand of cream
- If product data changes, recipe still works
- Menu can check "do we have ML-CR-001?"



MENU SYSTEM

Generation Logic

1. Scan current Inventory
2. Prioritize items:
 - a) Expiring Soon (HIGH)
 - b) Opened items (HIGH)
 - c) Old items (MEDIUM)
 - d) Regular items (LOW)
3. Find recipes using these items
4. Select recipes for meals
5. Reserve products
6. Calculate shopping needed

Menu Output

```
{
  "menuId": "MNU-20240115-001",
  "meals": [
    { "mealType": "breakfast", "recipes": ["RCP-003"] },
    { "mealType": "lunch", "recipes": ["RCP-005"] },
    { "mealType": "dinner", "recipes": ["RCP-001"] }
  ],
  "reservedInventory": [ ... ],
  "shoppingNeeded": [ ... ]
}
```



SHOPPING SYSTEM

Sources of Shopping Items

1. MENU SHORTAGES
Ingredients needed for selected menu
2. LOW STOCK
Items below Product.minimumStock

3. EXPIRING SOON (optional)
Use these items, buy similar

Shopping Item Priority

Priority 1 → Menu essential items
Priority 2 → Low stock replenishment
Priority 3 → Expiring items

PREP SYSTEM

What is Prep?

Prepared dishes or semi-finished products:

Examples:

- Chopped vegetables
- Marinated ingredients
- Half-cooked dishes
- Frozen batches

Storage Durations

Refrigerator: 1-5 days
Freezer: 1-3 months
Pantry: 1-2 weeks

Prep Structure

```
{  
  "prepId": "PREP-001",  
  "name": "Маринованные помидоры",  
  "productLinks": [  
    { "productId": "VG-TM-001", "quantity": 2000 }  
  ],  
  "portions": 8,  
  "storageLocation": "Freezer",  
  "shelfLife": 90  
}
```

API QUICK REFERENCE

Product Master

GET	/api/products	→ All products
GET	/api/products/:productId	→ Single product
POST	/api/products	→ Create product
PATCH	/api/products/:productId	→ Update product
GET	/api/products/search/:query	→ Search products

Inventory

GET	/api/inventory	→ All items
GET	/api/inventory/:inventoryId	→ Single item
POST	/api/inventory	→ Add to inventory
PATCH	/api/inventory/:inventoryId	→ Update item
GET	/api/inventory/product/:productId	→ All items of product
GET	/api/inventory/expiring?days=2	→ Expiring items

Receipts

POST	/api/receipts/scan	→ Upload & process receipt
GET	/api/receipts	→ All receipts
GET	/api/receipts/:receiptId	→ Receipt details

Menu & Shopping

POST	/api/menu/generate	→ Generate today's menu
GET	/api/shopping/list	→ Current shopping list

DATABASE KEYS

Product Master Index

PRIMARY KEY: product_id
FOREIGN KEY: category_code → categories(code)
INDEX: category_code, active, supplier

Inventory Index

PRIMARY KEY: inventory_id

FOREIGN KEY: product_id → products(product_id)

INDEX: product_id, location, status, expiration_date

KEY CONCEPTS

System Flow

```
PRODUCT MASTER (Root)
  ↓ (via productId)
  ├── INVENTORY (current stock)
  │   ├── MENU (what can we cook?)
  │   ├── SHOPPING (what we need)
  │   └── PREP (stored dishes)
  │
  ├── RECIPES (uses products)
  │   └── MENU (which recipes available?)
  │
  └── RECEIPTS (update prices)
```

The Three Laws

- 1** Product ID Never Changes
If you need to change it, you're doing it wrong
 - 2** Inventory ≠ Product Master
Separate concerns, separate tables
 - 3** Everything Connects via Product ID
No copies, no duplicates, single source of truth
-

DEVELOPMENT CHECKLIST

Phase 1: Product Master

- Create categories table
- Create products table
- Seed initial products
- Validate productId format
- API endpoints (CRUD)
- Frontend viewer

Phase 2: Inventory

- Create inventory table
- Link to Product Master (productId)
- Status management
- Expiration tracking
- API endpoints
- Inventory manager UI

Phase 3: Receipts

- OCR integration
- Dutch→Russian translation
- Product matching
- Auto-add to inventory
- Currency conversion
- Receipt upload UI

Phase 4: Recipes + Menu

- Recipes table (references productId)
- Recipe editor
- Menu generation algorithm
- Menu visualization
- Shopping list creation

Phase 5: Shopping

- Shopping list generation
- Shopping list UI
- Item checking
- Price tracking

Phase 6: Prep System

- Prep items table
- Prep storage tracking
- Portion management

Phase 7: Analytics

- Usage statistics
- Cost analysis
- Waste tracking
- Dashboard

UNIT REFERENCE

Weight

g = gram
kg = kilogram (1000g)
oz = ounce (≈28g)
lb = pound (≈454g)

Volume

ml = milliliter
l = liter (1000ml)
fl_oz = fluid ounce (≈30ml)
cup = cup (≈240ml)
tbsp = tablespoon (≈15ml)
tsp = teaspoon (≈5ml)

Count

piece = einzelne Stück
dozen = 12 pieces
bunch = пучок

IMMUTABILITY RULES

Product Master



CAN CHANGE:

- russianName, englishName, originalName
- category, subcategory
- baseUnit
- averagePrice
- supplier
- active status



CANNOT CHANGE:

- productId (NEVER)

Inventory



CAN CHANGE:

- quantity (use/add)
- location (move)
- status
- notes



CANNOT CHANGE:

- inventoryId (after creation)
 - productId (never reassign)
 - purchaseDate (historical)
-



TIPS & TRICKS

Finding duplicate product IDs

```
SELECT productId, COUNT(*)  
FROM products  
GROUP BY productId  
HAVING COUNT(*) > 1;
```

Checking inventory health

```
SELECT status, COUNT(*)  
FROM inventory  
GROUP BY status;
```

Finding expired items

```
SELECT * FROM inventory  
WHERE expiration_date < CURRENT_DATE;
```

Low stock alert

```
SELECT p.product_id, p.russian_name, i.quantity, p.minimum_stock  
FROM products p  
JOIN inventory i ON p.product_id = i.product_id  
WHERE i.quantity < p.minimum_stock;
```

TROUBLESHOOTING

Product not found in menu generation?

1. Check productId in recipe → productId in product master
2. Verify inventory has that product
3. Check inventory quantity >= recipe requirement

Receipt items not matching?

1. Check Dutch→Russian dictionary
2. Verify original product name in receipt
3. Check confidence score (should be > 0.8)

Inventory not updating from receipt?

1. Verify receipt processing completed
 2. Check inventory_id was created
 3. Verify product location is set
 4. Check expiration_date calculation
-



METRICS TO TRACK

Product Master:

- Total products: 100+
- Categories: 6
- Active products: 95%

Inventory:

- Total items: varies
- Fresh: ?%
- Opened: ?%
- Expiring Soon: <5%
- Expired: 0%

Menu:

- Average recipes per menu: 3
- Average shopping items: 8
- Waste reduction: target 30%

Shopping:

- Average cost: €15-20/day
 - Items per week: 20-30
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Remember: Product ID is immutable. Everything else flows from it. ✨